

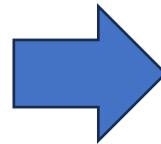
Introduction to the NGS data and data analysis

Dr. Samart Wanchana
BIOTEC, NSTDA

Where the NGS data come from?



170 coconut accessions



Next-generation sequencing



F - forward, R1

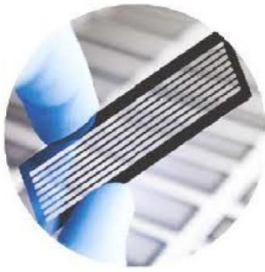
```
@M00218:49:000000000-A7RDL:1:1101:15401:1337 1:N:0:5
ACCCCTTTCCAATTGCGGCAGTTGTGTTTCCGGCAGCATCCAGAGCATCAGTTCTCTCCCGAATCTGTGGCTCATGCCGCCATTTCAGCAATTCCTCC
+
AABBBBFFFFFGGFGCGGGFGBBEFGGCEGGGHFFHHHHHEHFFHGHFHHGHFHGECFAFHF@GGHFGBEBEGE/EEFHHHHHHHGHFHHHHHH
@M00218:49:000000000-A7RDL:1:1101:15144:1344 1:N:0:5
TACGAATCCAAACATGGTCTGGTAGGTAAACGGATTAAAGTGGCAGTAGCATCAGCGGCGCTACGGACCACACAGCATTGTACTACTACATTACCCAAA
+
ABCCCCCFFFGGGGGGGGGHHFABGFHHGGGEFHHHGGFHGGGHHHHHHHHHHFCCGGFGGGGGG/@GHHGGFHHHHHFGHHHHHHHHHHHHHHHHHH
```

R - reverse, R2

```
@M00218:49:000000000-A7RDL:1:1101:15401:1337 2:N:0:5
AAATCAGTCATCATCCCTATTTTCTATTGCTTTAGCATTTCCTCAGCTTCAGCCTGCTGCCATGTATGGTGTGCTGGCTGCTCTTGGAAATGCT
+
BBBBBFFDFFGGGGFFGGCGHHHGHGBD5GGHHFGHFBGHHB5G33BFHHEFBGGHHGHDFCHHHGHDFHEGGGBDDF32BGCC3FHFHFB5G3D5
@M00218:49:000000000-A7RDL:1:1101:15144:1344 2:N:0:5
CGAGGAGGGCGACTACGAGGACGAAGAGGAGCAGTCCCCGAAGACAGTAATCTCGCCGAGTTCTCTCGGATCCCATCTTGTAGTGTGCGCTGTTGTG
+
BB?ABDBBBBGGGCGCEGGG?EGHGGGHHHHHHHGGCGGGBE?EGFHHHDHGGGGEE>FHHHHHFFGHGGHGHHHHHHHBGBHFFHGGEHHGHG
```

NSG data (fastq)

Illumina files: fastq



Sequence ID

Sequence

@HWI-ST330_0106:4:1:2643:2862#CTTGTA/1

CTTGACAAAGGGTGCAAGGCAGTTAGTGGTGCAAGATGCATTGCTGATGATGGGTTCATCAGGGCTGTAATCATA

+

ggggggggggggdgggeggfggggffafddgdgdgeggggggggggdggdggggceeedeegggg_eYdc_dcac → **Quality Scores**

@HWI-ST330_0106:4:1:2613:2891#CTTGTA/1

CGTGTCTTAAGGAGGCACCAAACAATATAAAGCTACAGATGGCGTCCTTGGTTTTTAATTTTAAGTTGGGGGACT

+

gggggggggggggegggdgggggegggffdggggggggggggggggggggggggfgggeeegggfgcgggfggge^eg

Different scoring versions

```
SSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSS.....  
...XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX.....  
.....IIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIII.....  
.....JJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJ.....  
LLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLL.....  
!"#$%&'()*+,-./0123456789:;<=?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~  
|                                     |  
33                               59   64       73                                104                             126  
0.....26...31.....40  
          -5...0.....9.....40  
                0.....9.....40  
                    3....9.....40  
0.2.....26...31.....41
```

- S - Sanger Phred+33, raw reads typically (0, 40)
X - Solexa Solexa+64, raw reads typically (-5, 40)
I - Illumina 1.3+ Phred+64, raw reads typically (0, 40)
J - Illumina 1.5+ Phred+64, raw reads typically (3, 40)
with 0=unused, 1=unused, 2=Read Segment Quality Control Indicator (bold)
(Note: See discussion above).
L - Illumina 1.8+ Phred+33, raw reads typically (0, 41)

https://en.wikipedia.org/wiki/FASTQ_format

Base Qualities

$$(\text{probability of error}) = 10^{-(\text{phred score}) / 10}$$

Phred quality score	Probability that the base is called wrong	Accuracy of the base call
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1,000	99.9%
40	1 in 10,000	99.99%
50	1 in 100,000	99.999%

Read alignment

Individual A



Individual B



ATGATAGCATCGTCGGGTGCTGCTCAATAATAGTGCCGTATCATGCTGGTGTATAATCGCCGCATGACATGATCAATGG

CAATAAAAGTGCCGTATCATGCTGGTGTACAATCGCCGCA

CGTATCATGCTGGTGTACAATCGCCGCATGACATGATCAATGG

TGTCTGCTCAATAAAAGTGCCGTATCATGCTGGTGTACAATC

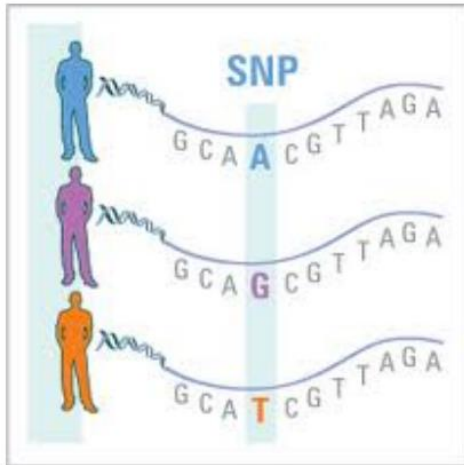
ATCGTCGGGTGCTGCTCAATAAAAGTGCCGTATCATG--GGTGTATAA

CTCAATAAGAGTGCCGTATCATG--GGTGTATAATCGCCGCA

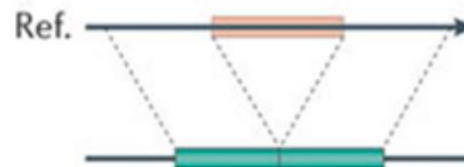
GTTATAATCGCCGCATGACATGATCAATGG

To measure variation.

Types of variants



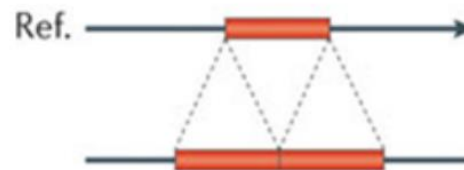
Deletion



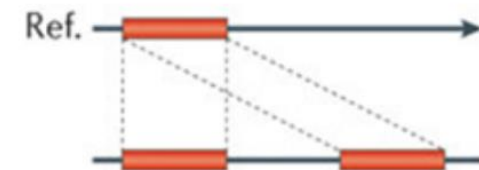
Novel sequence insertion



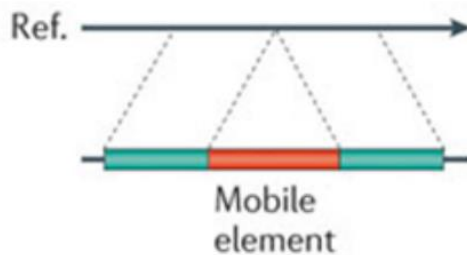
Tandem duplication



Interspersed duplication



Mobile-element insertion



Inversion

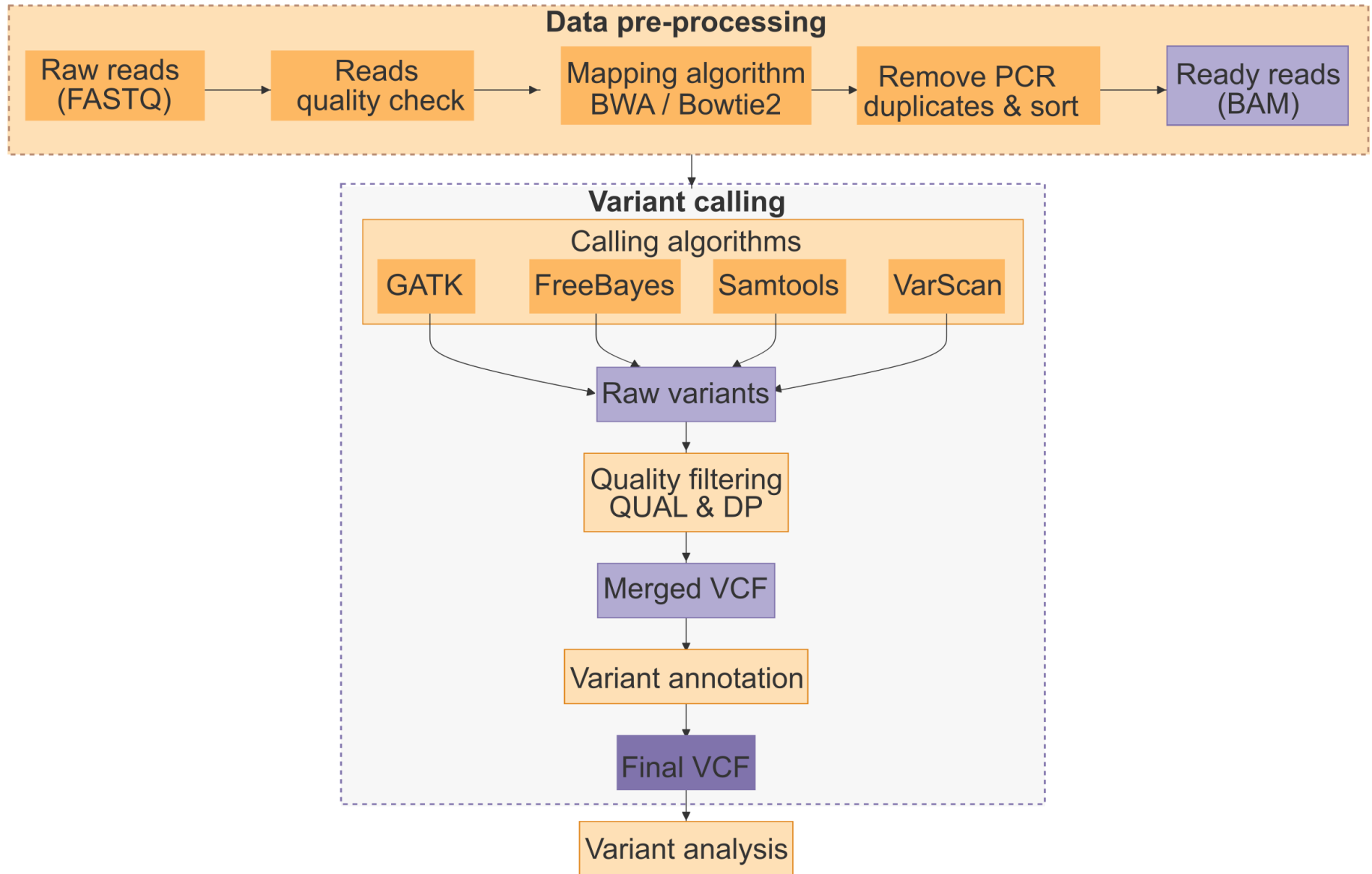


Translocation



Adapted from Alkan et al, Nature Reviews Genetics 2011

NGS data processing and Variant calling



Variant call format (VCF)

Example

VCF header

```
##fileformat=VCFv4.0
##fileDate=20100707
##source=VCFtools
##reference=NCBI36
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality (phred score)">
##FORMAT=<ID=GL,Number=3,Type=Float,Description="Likelihoods for RR,RA,AA genotypes (R=ref,A=alt)">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##ALT=<ID=DEL,Description="Deletion">
##INFO=<ID=SVTYPE,Number=1,Type=String,Description="Type of structural variant">
##INFO=<ID=END,Number=1,Type=Integer,Description="End position of the variant">
```

Mandatory header lines

Optional header lines (meta-data about the annotations in the VCF body)

Body

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	SAMPLE1	SAMPLE2
1	1	.	ACG	A,AT	.	PASS	.	GT:DP	1/2:13	0/0:29
1	2	rs1	C	T,CT	.	PASS	H2;AA=T	GT:GQ	0 1:100	2/2:70
1	5	.	A	G	.	PASS	.	GT:GQ	1 0:77	1/1:95
1	100	.	T		.	PASS	SVTYPE=DEL;END=300	GT:GQ:DP	1/1:12:3	0/0:20

Reference alleles (GT=0)

Alternate alleles (GT>0 is an index to the ALT column)

Deletion

SNP

Large SV

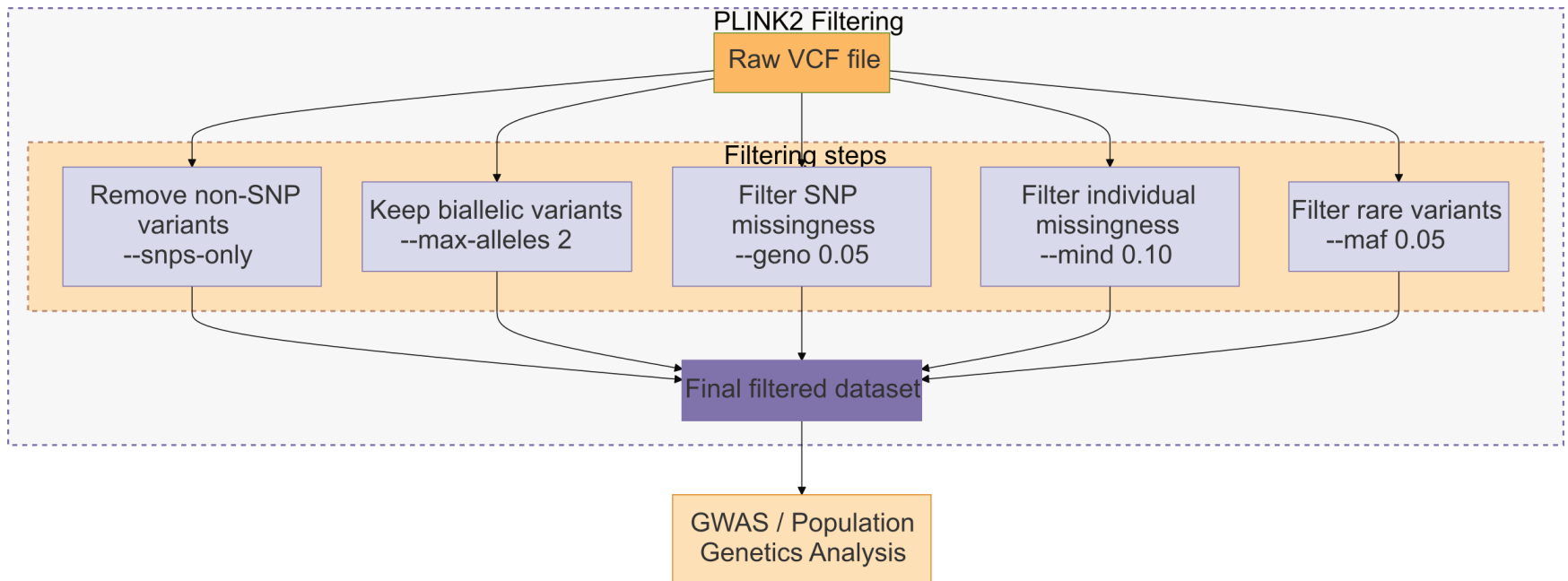
Insertion

Other event

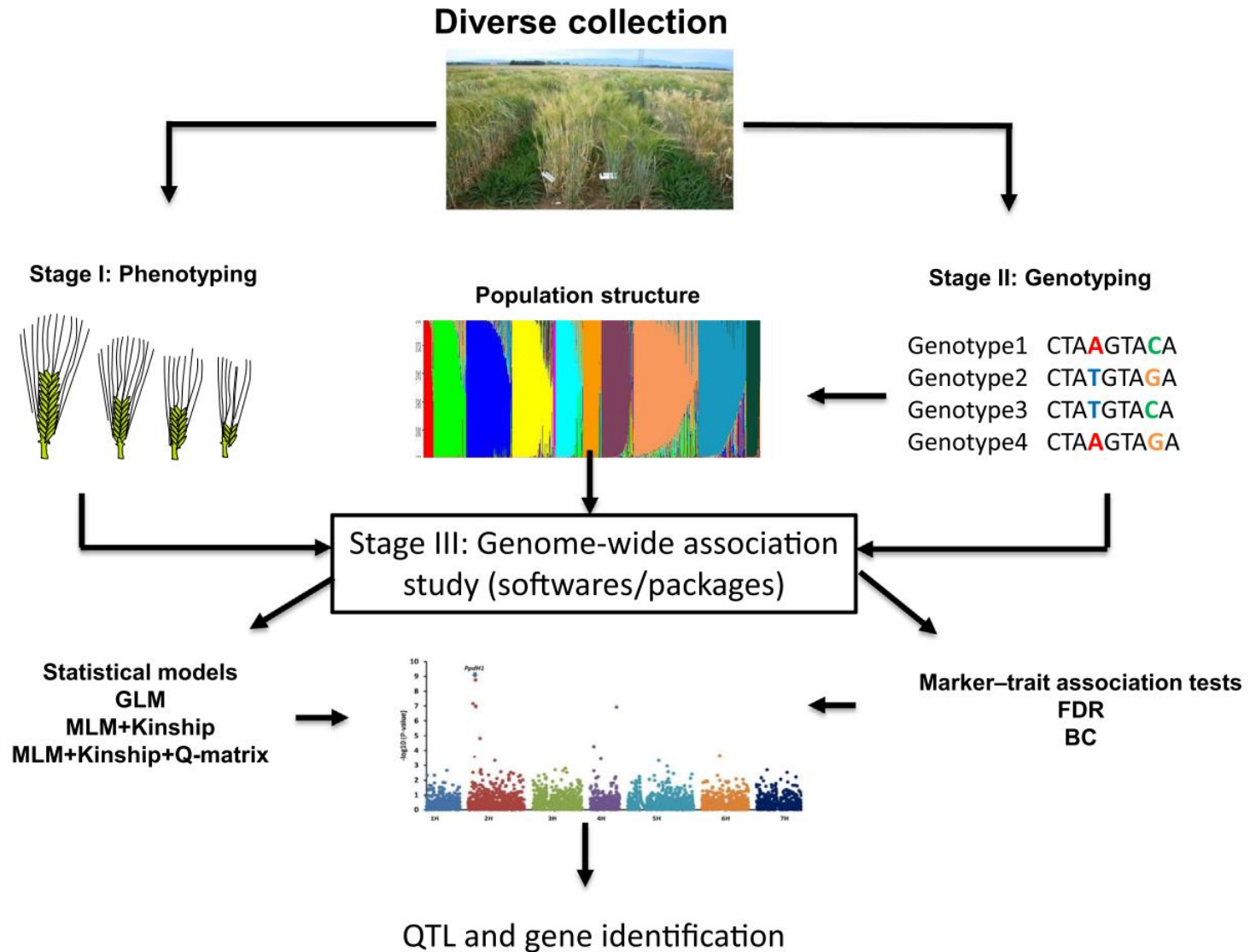
Phased data (G and C above are on the same chromosome)

<http://vcftools.sourceforge.net/specs.html> ... VCF poster

SNP Filtering



Genome-wide association study



Trait to analyze: Fruit Color



Acc.	Color	Color Code
FF68_CN1	Green	1
FF68_CN2	Green	1
FF68_CN3	Brown	3
FF68_CN4	Yellow	5
FF68_CN5	Yellow	5
FF68_CN6	Green	1
FF68_CN7	Green	1
FF68_CN8	Green	1
FF68_CN9	Green	1
FF68_CN10	Green	1
FF68_CN11	Green	1
FF68_CN12	Green	1
FF68_CN13	Yellowish green	2
FF68_CN14	Yellow	5
FF68_CN15	Green	1
FF68_CN16	Green	1
FF68_CN17	Green	1
FF68_CN18	Green	1
FF68_CN20	Green	1
FF68_CN21	Green	1
FF68_CN22	Green	1
FF68_CN23	Yellowish green	2
FF68_CN24	Green	1
FF68_CN25	Green	1
FF68_CN26	Green	1
FF68_CN27	Green	1
FF68_CN28	Green	1
FF68_CN29	Green	1
FF68_CN30	Green	1
FF68_CN31	Green	1
FF68_CN32	Green	1
FF68_CN33	Green	1
FF68_CN34	Green	1
FF68_CN35	Green	1